

# Problem Set 1

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**Problem 1:**

Goofus, a MAT102 student, has asked you to critique his proof.

**Proposition.** Let  $n \in \mathbb{Z}$ . Then  $n$  is odd if and only if  $n^2$  is odd.

*Proof.*  $n$  odd  $\implies \exists k, n = 2k + 1 \implies n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2l + 1 \implies n^2$  odd.

$n$  even  $\implies n = 2k \implies n^2 = (2k)^2 = 4k^2 = 2(2k^2) = 2l \therefore n^2$  even.

Hence proved. □

Goofus' proof is logically correct, but very poorly communicated. Unfortunately, Goofus would get a score of 0/10 for submitting this proof in MAT301. Nice try, Goofus!

(a) Point out five things in Goofus' proof that need improvement.

**Solution:**

(b) Write your own proof of this proposition, focusing on improving the communication issues you noticed in Goofus' proof.

**Solution:**

**Problem 2:**

Consider the following binary operation on  $\mathbb{Q}$ :

$$x \otimes y := x + y + xy$$

For example,  $2 \otimes 3 = 11$ .

(a) Prove that  $\otimes$  is associative.

**Solution:**

Let  $a, b$ , and  $c \in \mathbb{Q}$  be arbitrary. To prove associativity, we must show that  $(a \otimes b) \otimes c = a \otimes (b \otimes c)$ . By simply using the given definition of  $\otimes$  and basic facts related to multiplication and addition on  $\mathbb{Q}$ , we can indeed see that:

$$\begin{aligned}
 (a \otimes b) \otimes c &= (a + b + ab) \otimes c && \text{(By definition of } \otimes \text{)} \\
 &= a + b + ab + c + (a + b + ab)c && \text{(By definition of } \otimes \text{)} \\
 &= a + b + ab + c + ac + bc + abc && \text{(By distributivity of multiplication over addition on } \mathbb{Q} \text{)} \\
 &= a + ab + ac + abc + b + c + bc && \text{(By commutativity of addition on } \mathbb{Q} \text{)} \\
 &= a + ab + ac + abc + (b \otimes c) && \text{(By definition of } \otimes \text{)} \\
 &= a + a(b + c + bc) + (b \otimes c) && \text{(By distributivity of multiplication over addition on } \mathbb{Q} \text{)} \\
 &= a + a(b \otimes c) + (b \otimes c) && \text{(By definition of } \otimes \text{)} \\
 &= a + (b \otimes c) + a(b \otimes c) && \text{(By commutativity of addition on } \mathbb{Q} \text{)} \\
 &= a \otimes (b \otimes c) && \text{(By definition of } \otimes \text{)}
 \end{aligned}$$

And so, since  $(a \otimes b) \otimes c = a \otimes (b \otimes c)$ ,  $\otimes$  is associative over  $\mathbb{Q}$ .

(b) Prove that  $\otimes$  is commutative.

**Solution:**

Let  $a, b \in \mathbb{Q}$  be arbitrary. To prove commutativity, we must show that  $a \otimes b = b \otimes a$ , by a similar process as before, we can see that:

$$\begin{aligned}
 a \otimes b &= a + b + ab && \text{(By definition of } \otimes \text{)} \\
 &= b + a + ab && \text{(By commutativity of addition on } \mathbb{Q} \text{)} \\
 &= b + a + ba && \text{(By commutativity of multiplication on } \mathbb{Q} \text{)} \\
 &= b \otimes a && \text{(By definition of } \otimes \text{)}
 \end{aligned}$$

Thus, since  $a \otimes b = b \otimes a$ ,  $\otimes$  is indeed commutative over  $\mathbb{Q}$ .

(c) Find, with a proof, the identity element for  $\otimes$ .

**Solution:**

The identity element for  $\otimes$  in  $\mathbb{Q}$  is 0. To prove that 0 is really  $\otimes$ 's identity element, we need to show that for arbitrary  $a \in \mathbb{Q}$ ,  $a \otimes 0 = 0 \otimes a = a$ . Letting  $a$  be any such element in  $\mathbb{Q}$  then, expanding gives:

$$\begin{aligned}
\boxed{a \otimes 0} &= a + 0 + a(0) && \text{(By definition of } \otimes \text{)} \\
&= a + 0 + 0 && \text{(By property of multiplication by zero)} \\
&= \boxed{a} && \text{(Since 0 is the additive identity on } \mathbb{Q} \text{)} \\
&= 0 + a + 0 && \text{(Since 0 is the additive identity on } \mathbb{Q} \text{)} \\
&= 0 + a + 0(a) && \text{(By property of multiplication by zero)} \\
&= \boxed{0 \otimes a} && \text{(By definition of } \otimes \text{)}
\end{aligned}$$

Therefore, since 0 satisfies  $a \otimes 0 = 0 \otimes a = a$  for arbitrary  $a \in \mathbb{Q}$ , 0 is the identity element of  $\otimes$  over  $\mathbb{Q}$ .

(d) Find, with proof, the inverse of  $\frac{5}{6}$ .

**Solution:**

The inverse of  $\frac{5}{6}$  is  $\frac{-5}{11}$ . To prove that this is indeed the case, we need to show that for  $\frac{5}{6} \otimes \frac{-5}{11} = \frac{-5}{11} \otimes \frac{5}{6} = 0$ , since we previously determined 0 to be the identity element of  $\otimes$  over  $\mathbb{Q}$ . Expanding out, we find that:

$$\begin{aligned}
\boxed{\frac{5}{6} \otimes \frac{-5}{11}} &= \frac{5}{6} - \frac{5}{11} + \left(\frac{5}{6}\right) \left(\frac{-5}{11}\right) && \text{(By definition of } \otimes \text{)} \\
&= \frac{55}{66} - \frac{30}{66} - \frac{25}{66} && \text{(Simplifying)} \\
&= \boxed{0} && \text{(Simplifying)} \\
&= -\frac{30}{66} + \frac{55}{66} - \frac{25}{66} && \text{(Expanding)} \\
&= -\frac{-5}{11} + \frac{5}{6} + \left(\frac{-5}{11}\right) \left(\frac{5}{6}\right) && \text{(Expanding)} \\
&= \boxed{\frac{5}{6} \otimes \frac{-5}{11}} && \text{(By definition of } \otimes \text{)}
\end{aligned}$$

And so,  $-\frac{5}{11}$  is the inverse of  $\frac{5}{6}$  under  $\otimes$  in  $\mathbb{Q}$ .

(e) Describe all rational numbers that are invertible with respect to  $\otimes$ . In fact, given a rational number  $x \in \mathbb{Q}$ , find a formula for the inverse of  $x$  with respect to  $\otimes$ .

**Solution:**

All rational numbers, except  $-1$ , are invertible.

To see this clearly, I will proceed by a direct proof. Letting  $a \in \mathbb{Q}$  be arbitrary, we know definitionally that the inverse of  $a$  is an element  $a^{-1} \in \mathbb{Q}$  satisfying  $a \otimes a^{-1} = a^{-1} \otimes a = 0$ , since 0 was determined earlier to be the identity element of  $\otimes$  over  $\mathbb{Q}$ . By definition of  $\otimes$ , we therefore can see that  $a \otimes a^{-1} = 0$  yields:

$$\begin{aligned}
& a \otimes a^{-1} = 0 && \text{(By definition of inverse)} \\
\implies & a + a^{-1} + aa^{-1} = 0 && \text{(By definition of } \otimes \text{)} \\
\implies & a + (1 + a)a^{-1} = 0 && \text{(By distributivity of multiplication over addition on } \mathbb{Q} \text{)} \\
\implies & (1 + a)a^{-1} = -a && \text{(Moving } a \text{ to the right-hand side)} \\
\implies & a^{-1} = -\frac{a}{1+a} && \text{(Moving } a \text{ to the right-hand side)}
\end{aligned}$$

Notice however that the final equality we arrived at is undefined in the case that  $a = -1$ , meaning that  $-1 \in \mathbb{Q}$  cannot possibly have an inverse under  $\otimes$ . The equality however is perfectly valid for all other numbers in  $\mathbb{Q}$ . Thus, every element in  $\mathbb{Q} \setminus \{-1\}$  has an inverse under  $\otimes$ , and for any  $a \in (\mathbb{Q} \setminus \{-1\})$ , its inverse is given by  $a^{-1} = -\frac{a}{1+a}$ .

**Problem 3:**

Let  $2 \leq n$  and let  $u \in \mathbb{Z}_n$ . Recall that  $u$  is a “unit modulo  $n$ ” if it is multiplicatively invertible. Thus, there exists  $v \in \mathbb{Z}_n$  such that  $uv \equiv 1 \pmod{n}$ . Let  $\mathbb{Z}_n^\times$  denote the set of all unit modules of  $n$ ; For example,  $\mathbb{Z}_9^\times = \{1, 2, 4, 5, 7, 8\}$ .

(a) Write down the multiplication table for the group  $\mathbb{Z}_{36}^\times$ .

**Solution:**

First, we must actually determine  $\mathbb{Z}_{36}^\times$ .